

HOMEWORK 1

CMU 10-753: DL AND GENAI THEORY

OUT: Thursday, Sept 3th, 2026

DUE: Thursday, Sept 24th, 2026, 11:59pm

START HERE: Instructions

- **Collaboration policy:** Collaboration on solving the homework is allowed, after you have thought about the problems on your own. To remind you, many questions in this HW have solutions that are very easy to find online (and many are from previous versions of this course). It is also OK to get clarification (but not solutions) from books or online resources, again after you have thought about the problems on your own. There are two requirements: first, cite your collaborators fully and completely (e.g., “Jane explained to me what is asked in Question 2.1”). Second, write your solution *independently*: close the book and all of your notes, and send collaborators out of the room, so that the solution comes from you only.
- **Submitting your work:**
 - **Gradescope:** For the written problems such as short answer, multiple choice, derivations, proofs, or plots, we will be using the Gradescope. The best way to format your homework is by using the Latex template released in the handout and writing your solutions in Latex. However, submissions can be handwritten onto the template, but should be labeled and clearly legible. If your writing is not legible, you will not be awarded marks.
Regrade requests can be made after the homework grades are released, however this gives the TA the opportunity to regrade your entire paper, meaning if additional mistakes are found then points will be deducted.
 - **Programming:** You should submit all code used to solve the programming aspect of the homework to the corresponding ‘Programming’ submission slot on Gradescope. If you do not do this, you will not get any credit for any of the programming section irrespective of the plots and values submitted to the ‘Written’ submission slot.

1 α_t and β_t Schedules in Denosing Diffusion - Saumya

Recall the definition of the forward diffusion from class, where given a schedule $\beta_1, \dots, \beta_T$, the distribution of the T th time step given a sample at the 0th time step is given by:

$$q(x_T|x_0) = \mathcal{N}\left(x_T; \sqrt{\prod_{t=1}^T (1 - \beta_t)} x_0, (1 - (\prod_{t=1}^T (1 - \beta_t))) \mathbf{I}\right).$$

Alternatively, we define $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ to get:

$$q(x_T|x_0) = \mathcal{N}(x_T; \sqrt{\bar{\alpha}_T} x_0, (1 - \bar{\alpha}_T) \mathbf{I}).$$

The Wasserstein W_2 distance between two Gaussians admits a clean closed-form expression as given below:

$$W_2^2(\mathcal{N}(\mu_1, \Sigma_1), \mathcal{N}(\mu_2, \Sigma_2)) = \|\mu_1 - \mu_2\|^2 + \text{Tr}(\Sigma_1 + \Sigma_2 - 2(\Sigma_1^{1/2} \Sigma_2 \Sigma_1^{1/2})^{1/2}).$$

1.1 Wasserstein Distance as a Function of $\bar{\alpha}_T$ (5 points)

How does the W_2 distance between the final step of the diffusion process ($q(x_T|x_0)$) and random noise ($\mathcal{N}(0, \mathbf{I})$) depend on $\bar{\alpha}_T$? How does the W_2 distance change as $\bar{\alpha}_T \rightarrow 0$?

1.2 Linear Schedule vs Constant Schedule (5 points)

Consider the linear schedule given by $\beta_t = \beta_{\min} + \frac{t-1}{T-1}(\beta_{\max} - \beta_{\min})$ for $\beta_{\min} < \beta_{\max}$. Use the bound $\log(1-x) \geq -\frac{x}{1-x}$ to derive a lower bound on $\bar{\alpha}_T$. Assume $\beta_{\max} \ll 1$ to show that, to leading order in $\beta_{\max}$, $\bar{\alpha}_T$ for the linear schedule decays at the same rate as $\bar{\alpha}_T$ for a constant schedule with $\beta_t = (\beta_{\max} + \beta_{\min})/2$.

1.3 Cosine Schedule (5 points)

Another commonly used schedule is the cosine schedule, which directly sets:

$$\bar{\alpha}_t = \frac{\cos^2\left(\frac{t/T + \pi}{s+1}\right)}{\cos^2\left(\frac{\pi}{s+1}\right)}.$$

Why does such a schedule help the forward diffusion be more “precise” at a fixed/limited budget T ?

2 Stochastic Convergence - Ibrahim

Let $\mathbb{P}$ be a probability space, and let $X, X_1, X_2, \dots$ be real-valued random variables defined on this space.

Convergence in L^p . For $p \geq 1$, we say $X_n \rightarrow X$ in L^p if

$$\mathbb{E}[|X_n - X|^p] \rightarrow 0.$$

In particular, L^1 convergence means $\mathbb{E}[|X_n - X|] \rightarrow 0$, and L^2 convergence means $\mathbb{E}[|X_n - X|^2] \rightarrow 0$.

Convergence in probability. We say $X_n \rightarrow X$ in probability if, for every $\epsilon > 0$,

$$\mathbb{P}(|X_n - X| > \epsilon) \rightarrow 0.$$

Convergence in distribution. We say $X_n \rightarrow X$ in distribution if $F_{X_n}(x) \rightarrow F_X(x)$ at every point x where F_X is continuous.

Hints. You may want to use these properties.

Markov's inequality If $Y \geq 0$, then for every $a > 0$,

$$\mathbb{P}(Y \geq a) \leq \frac{\mathbb{E}[Y]}{a}.$$

Hölder's inequality. For random variables U, V and conjugate exponents $p, q > 1$ satisfying $\frac{1}{p} + \frac{1}{q} = 1$,

$$|\mathbb{E}[UV]| \leq (\mathbb{E}[|U|^p])^{1/p} (\mathbb{E}[|V|^q])^{1/q}.$$

2.1 Convergence in L_2 , L_1 , probability, and distribution (15 points)

1. [5pts] Suppose that $X_n \rightarrow X$ in L^2 , i.e. $\mathbb{E}[(X_n - X)^2] \rightarrow 0$. Prove that $X_n \rightarrow X$ in probability:

$$X_n \xrightarrow{L^2} X \Rightarrow X_n \xrightarrow{P} X.$$

2. [5pts] Let $p > q \geq 1$. Suppose that $X_n \rightarrow X$ in L^p , i.e. $\mathbb{E}[|X_n - X|^p] \rightarrow 0$. Prove that $X_n \rightarrow X$ in L^q :

$$X_n \xrightarrow{L^p} X \Rightarrow X_n \xrightarrow{L^q} X.$$

(Hint: apply Hölder's inequality to $|X_n - X|^q \cdot 1$.)

3. [5pts] Using the previous two questions (with $p = 2, q = 1$), prove the following chain:

$$X_n \xrightarrow{L^2} X \Rightarrow X_n \xrightarrow{L^1} X \Rightarrow X_n \xrightarrow{P} X \Rightarrow X_n \xrightarrow{d} X.$$

You may use, without proof, results from previous questions and the fact that $X_n \xrightarrow{P} X \Rightarrow X_n \xrightarrow{d} X$.

2.2 Counterexamples (15 points)

Consider the chain $X_n \xrightarrow{L^2} X \Rightarrow X_n \xrightarrow{L^1} X \Rightarrow X_n \xrightarrow{P} X \Rightarrow X_n \xrightarrow{d} X$. Determine whether each of the three reverse implications is true. For every false implication, give a counterexample.

1. [5pts] Does L^1 convergence imply L^2 convergence?
2. [5pts] Does convergence in probability imply L^1 convergence?
3. [5pts] Does convergence in distribution imply convergence in probability?

2.3 Convergence in Generative Models (5 points)

The Portmanteau theorem states that

$$X_n \xrightarrow{d} X$$

if and only if

$$\mathbb{E}[f(X_n)] \longrightarrow \mathbb{E}[f(X)]$$

for every bounded continuous function f .

Suppose P_n is the distribution generated by a sequence of generative models and

$$P_n \Rightarrow P_{\text{data}},$$

i.e., P_n converges weakly to the data distribution.

What can you conclude about

$$\mathbb{E}_{X \sim P_n}[f(X)]$$

for a bounded continuous statistic f ? What does this suggest about which evaluation metrics are guaranteed to converge under weak convergence? In particular, discuss Maximum Mean Discrepancy (MMD) with a bounded continuous kernel and contrast it with metrics based on unbounded statistics, such as FID.

3 One forward step at a time - Krish (15 points)

In lecture the forward process was defined one step at a time,

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I}\right), \quad (1)$$

with closed form

$$q(x_t \mid x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) \mathbf{I}\right). \quad (2)$$

This problem carries out the inductive step of that proof on t . Write the forward step (1) as

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} z_t, \quad z_t \sim \mathcal{N}(0, \mathbf{I}) \text{ independent of } x_{t-1},$$

and recall $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$, $\bar{\alpha}_0 = 1$. The base case $t = 0$ is the identity $x_0 = \sqrt{\bar{\alpha}_0} x_0$.

1. [5pts] Assume that (2) holds at time $t - 1$, that is, $x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} x_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon'$ with $\epsilon' \sim \mathcal{N}(0, \mathbf{I})$ independent of z_t . Substitute this into the forward step and show that $\mathbb{E}[x_t | x_0] = \sqrt{\bar{\alpha}_t} x_0$.
2. [5pts] Show that the covariance of x_t given x_0 is $(1 - \bar{\alpha}_t)\mathbf{I}$, and conclude that (2) holds at time t . *Hint:* a sum of independent Gaussian vectors is Gaussian, and its covariance is the sum of the covariances.
3. [5pts] *Final observation.* Assume the data are normalized so that $\mathbb{E}\|x_0\|^2 = d$, where d is the dimension of x_0 . Compute $\mathbb{E}\|x_t\|^2$ from the form $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$ and say in one sentence what happens to $q(x_T | x_0)$ when $\bar{\alpha}_T \rightarrow 0$. For the schedule $\beta_t = t \cdot 10^{-4}$, $t = 1, \dots, 100$, give $\bar{\alpha}_{100}$, $\sqrt{\bar{\alpha}_{100}}$ and $\sqrt{1 - \bar{\alpha}_{100}}$ to three decimals.

4 Covariance of the Wiener process - Krish (15 points)

Let $\{W_t\}_{t \geq 0}$ be a standard Wiener process as defined in lecture:

$$W_0 = 0, \quad W_t - W_s \sim \mathcal{N}(0, t - s) \text{ for } 0 \leq s < t, \quad (3)$$

and increments over disjoint intervals are independent.

In class we derived from this

$$\mathbb{E}[W_t] = 0, \quad \text{Var}(W_t) = t, \quad \text{Cov}(W_s, W_t) = \min(s, t) \text{ for all } s, t \geq 0. \quad (4)$$

This problem practises using these facts, and the bilinearity of covariance, to compute second moments of linear combinations of the values W_t .

1. [5pts] Let $0 \leq s < t$. Using only $\text{Cov}(W_s, W_t) = \min(s, t)$ and bilinearity, compute $\text{Var}(W_t - W_s)$ and check that the result agrees with the increment law in (3).
2. [5pts] Compute $\text{Var}(W_1 + W_2)$ and $\text{Cov}(W_1, W_3 - W_2)$. Explain the second answer in one sentence using the independent-increments property of (3).
3. [5pts] *Final observation.* For $0 < s < t$ compute the correlation coefficient

$$\text{Corr}(W_s, W_t) = \frac{\text{Cov}(W_s, W_t)}{\sqrt{\text{Var}(W_s) \text{Var}(W_t)}},$$

evaluate it for $s = 1$, $t = 4$, and state its limit as $t \rightarrow \infty$ with s fixed. What does the limit say about how the process remembers its early values?

5 Programming Question - Ibrahim (30 points)

Your task is to implement and train a denoising diffusion model on a simple 2D dataset. We provide a Jupyter notebook file with a few missing cells https://github.com/bapoczko/10753/blob/main/hw1_diffusion_2d_question.ipynb. Please follow the instructions in this notebook and complete the missing parts.

You will need to submit your completed Jupyter notebook. In this notebook, you will need to implement:

- Cosine beta schedule (5 points)
- Forward diffusion (10 points)
- Implement the noise-prediction training objective (5 points)
- Implement the reverse diffusion for sampling (10 points)

Important!

Important: Remember that you **MUST** submit all code used in this part to the Programming submission slot on Gradescope otherwise you will not get credit for this section.

Before generating any figures below or submitting, restart the kernel and run all cells top to bottom (Kernel/Runtime → Restart & Run All) so the notebook you submit reproduces cleanly from a blank state. Submit this restarted, fully-run notebook (.ipynb) along with your write-up.

As a proof-of-execution check, add the following figures directly below this question in this document (not just in your notebook). Each figure must have a caption that clearly indicates which figure is being shown. You will separately submit your completed Jupyter notebook (.ipynb) as well.

1. **Cosine beta schedule:** Run the cell that plots your implemented cosine beta schedule, and add the resulting figure here.
2. **Forward diffusion:** Run the provided 5-panel figure cell that plots your noising process, and add the resulting figure here.
3. **Training objective:** Run the cell that plots the training loss curve produced by your implemented training objective, and add the resulting figure here.
4. **Reverse diffusion / sampling:** Run the sampling cell that plots the "Generated samples" scatter plot, and add the resulting figure here.

6 Collaboration Questions

1. (a) Did you receive any help whatsoever from anyone in solving this assignment?
(b) If you answered ‘yes’, give full details (e.g. “Jane Doe explained to me what is asked in Question 3.4”)

2. (a) Did you give any help whatsoever to anyone in solving this assignment?
(b) If you answered ‘yes’, give full details (e.g. “I pointed Joe Smith to section 2.3 since he didn’t know how to proceed with Question 2”)

3. (a) Did you find or come across code that implements any part of this assignment?
(b) If you answered ‘yes’, give full details (book & page, URL & location within the page, etc.).